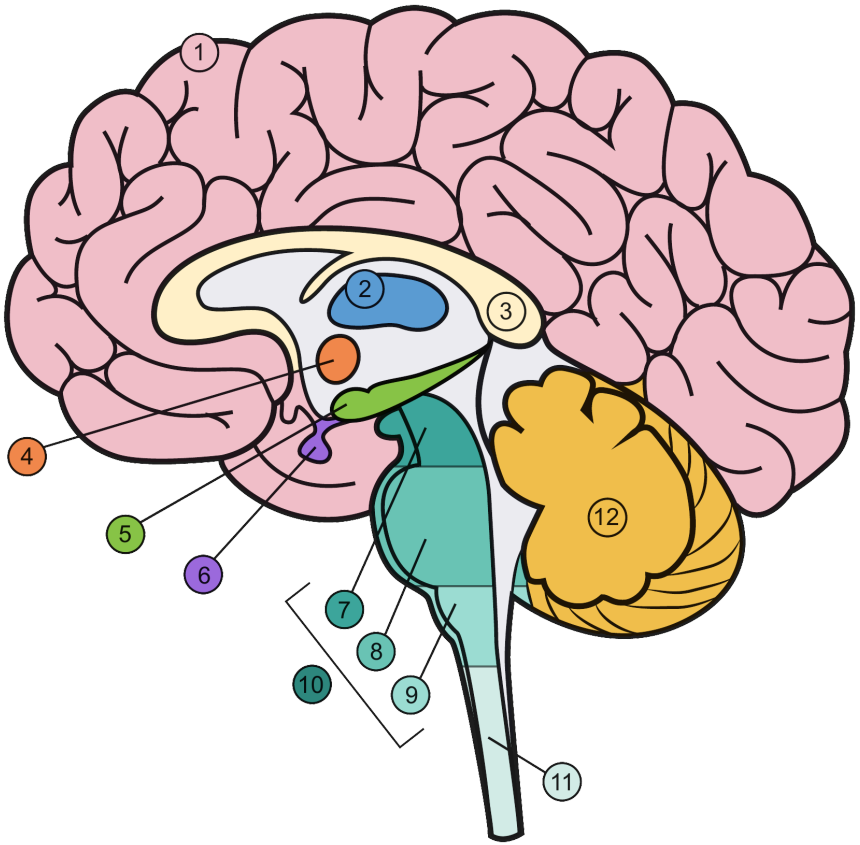
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Brain Anatomy — Reference Chart

MCB80x · midsagittal view · 12 structures, divisions, functions, and exam facts

DIAGRAM



Midsagittal section, anterior (front) to the left. Gray = ventricles / unlabeled tissue.

DEVELOPMENTAL DIVISIONS

Forebrain	Midbrain	Hindbrain	Other
Telencephalon: 1 3 5	Mesencephalon: 7	Pons 8 Medulla 9	Gland 6
Diencephalon: 2 4		Cerebellum 12	Cord 11

KEY

#	Structure	Function · exam fact
1	**Cerebral cortex** Telencephalon · gray matter	Six-layered outer sheet; perception, voluntary movement, language, planning. **FACT** Four lobes (frontal, parietal, temporal, occipital); gyri and sulci multiply surface area.
2	**Thalamus** Diencephalon	Relay hub sending every sense except smell on to the cortex. **FACT** LGN relays vision, MGN relays hearing.
3	**Corpus callosum** Telencephalon · white matter	~200 million axons linking left and right hemispheres. **FACT** Cut in "split-brain" surgery for severe epilepsy.
4	**Hypothalamus** Diencephalon	Homeostasis: temperature, hunger, thirst, circadian rhythm (SCN). **FACT** Makes oxytocin and ADH; releasing hormones steer the anterior pituitary.
5	**Hippocampus** Telencephalon · limbic	Forms new declarative and spatial memories; place cells. **FACT** Patient H.M.: bilateral removal → anterograde amnesia. LTP first shown here.
6	**Pituitary gland** Endocrine gland	Anterior lobe: GH, ACTH, TSH, LH/FSH, prolactin. **FACT** Posterior lobe is neural tissue — hypothalamic axon terminals releasing oxytocin and ADH.
7	**Midbrain** Mesencephalon	Colliculi: superior (visual orienting), inferior (auditory). **FACT** Substantia nigra dopamine loss → Parkinson's disease.
8	**Pons** Hindbrain · metencephalon	Relays cortex → cerebellum; helps set breathing rhythm and REM sleep. **FACT** Cranial nerves V–VIII exit at the pons and pontomedullary junction.
9	**Medulla oblongata** Hindbrain · myelencephalon	Heart rate, blood pressure, breathing, swallowing. **FACT** Corticospinal tracts cross here, so each hemisphere moves the opposite side.
10	**Brainstem** = 7 + 8 + 9	Links brain to spinal cord; reticular formation controls arousal. **FACT** Houses cranial nerve nuclei III–XII.
11	**Spinal cord** Central nervous system	Ascending sensory and descending motor tracts; spinal reflexes. **FACT** Dorsal root = sensory in, ventral root = motor out.
12	**Cerebellum** Hindbrain · metencephalon	Coordination, balance, timing, motor learning. **FACT** Holds ~80% of the brain's neurons; Purkinje cells are its sole output.

Personal MCB80x study chart — not an official HarvardX/edX publication. Line art: Ask A Biologist (askabiologist.asu.edu), CC BY-NC-SA; colorized.